Iteration: FOR, WHILE and REPEAT loops

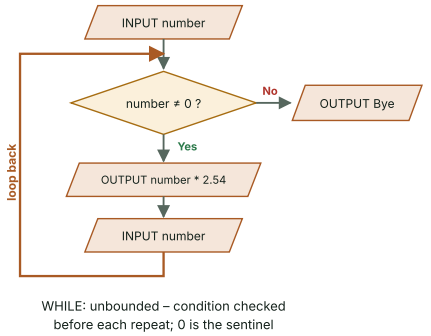
D16 FACT FILE

Specification: students should be able to

- Controlling the flow of a program through sequence, selection and iteration (iteration)

CCEA fact file: Programming Constructs 3

Key words: iteration, loop, FOR, WHILE, REPEAT UNTIL, bounded, unbounded, count-controlled, condition-controlled, running total, counter, sentinel, infinite loop



WHILE: unbounded – condition checked before each repeat; 0 is the sentinel

```
for (int i = 0; i < 5; i++)
```

0 1 2 3 4

FOR: bounded – repeats a set number of times. i < 5 gives 0 to 4, not 5.

do ... while (C#) and REPEAT...UNTIL check at the end: always run once.

Flowchart of a WHILE loop with a sentinel value, and a FOR loop counting

Types of iteration

Bounded iteration repeats a fixed number of times using a FOR loop. Unbounded iteration repeats while (or until) a condition holds: a WHILE loop tests its condition before each repetition, so it can run zero times; REPEAT...UNTIL tests at the end, so it always runs at least once. A sentinel value is a special input that signals the end of data, such as 0.

Common patterns

Initialise a running total or counter before the loop, update it inside, and use or output it after the loop ends. Make sure the loop variable changes inside a WHILE loop so the condition eventually becomes False; otherwise the result is an infinite loop.

T Key terms from the mark schemes

Iteration
Repeating a block of code (loops).
Credited: U4 2019 Q4, 2025 Q2

Infinite loop
A loop whose condition never becomes false, so it never ends.
Credited: U4 2025 Q2

C C# example

FOR loops

```
int total = 0;
for (int count = 1; count <= 5; count++)
{
    Console.Write("Mark: ");
    int mark = Convert.ToInt32(Console.ReadLine());
    total = total + mark;
}
Console.WriteLine("Average: " + (double)total / 5);
```

Exam focus

Loop output prediction (2024 Q2, 2025 Q2), fixing a faulty condition (2025 Q2(c)), running totals (2019 Q2(c), 2023 Q3(c)), WHILE loops with arrays (2024 Q7(c)(iii), 2025 Q8(b)(iii)). Examiners report pupils prefer FOR loops – practise WHILE loops too.

Unit 4 2025 Q8	[20]	Array properties, initialise, count, WHILE totals
Unit 4 2025 Q2	[12]	Loop output, nested loops, loop count, infinite loop, fixing a condition
Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2024 Q2	[3]	Tracing FOR and WHILE loops
Unit 4 2023 Q3	[12]	Data types; constants and a benefit; sales total and average algorithm

? Check yourself

1. Describe the difference between a FOR loop and a WHILE loop. [2]
2. What is output? total ← 0; FOR i ← 1 TO 4: total ← total + i * 2; NEXT i; OUTPUT total [2]
3. Explain what an infinite loop is and give one cause. [2]

Answers: 1. A FOR loop is bounded/count-controlled: it repeats a known/fixed number of times (1) / a WHILE loop is unbounded/condition-controlled: it repeats while a condition is True, checked before each repeat (so it may run zero times) (1) 2. Working: 2 + 4 + 6 + 8 (1) / output 20 (1) 3. An infinite loop never ends because its condition never becomes False (1) / cause, e.g. the loop variable is not changed inside the loop, or the condition uses the wrong comparison (1)